The Ch(e)at Code

Agent Patterns for Copilot Chat

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ISSUE #017 • JULY 21, 2026 • 🔧 PRACTICAL BUILD

Last issue, we named the Evaluation & Trust pattern — three layers of metrics, hallucination detection, regression testing, and user trust signals. This issue, we build the dashboard. By the end, you'll have a working analytics stack that answers two questions in real-time: "How many people used your agent yesterday?" and "Were the answers any good?"

This issue's build: A complete agent analytics stack using Copilot Studio built-in analytics, Application Insights custom telemetry, a Power BI dashboard, an automated regression alert flow, and a user feedback capture system.

🎮 AGENT SPOTLIGHT

The Agent Analytics Dashboard

Copilot Studio + Application Insights + Power BI + Power Automate

The scenario: A customer shipped their Copilot Studio agent six weeks ago. Leadership is asking for numbers. The agent owner opens the built-in analytics page and sees session counts — but nothing about answer quality, confidence trends, or user satisfaction. They can tell you the agent was used 200 times last week. They can't tell you whether those 200 interactions were helpful.

The solution: Layer five analytics components on top of an existing Copilot Studio agent. The built-in analytics handle adoption. Application Insights captures quality signals. Power BI visualizes both. Power Automate automates regression detection and feedback collection. Total build time: one focused session.

Why this stack: Every component is native to the Microsoft ecosystem — no third-party tools, no custom infrastructure. Copilot Studio has a built-in Application Insights connector. Power BI connects to App

Insights via the KQL data source. Power Automate orchestrates the regression tests and feedback flows. The customer likely already has licenses for all of it.

🧩 **PATTERN BREAKDOWN**

Building Agent Analytics in Copilot Studio

Five components: built-in analytics + App Insights telemetry + Power BI dashboard + regression alerts + user feedback

Why this matters: Issue #016 defined the three evaluation layers: usage, quality, and business metrics. This build implements all three with tools the customer already has. No excuses, no custom development — just configuration and flows.

END-TO-END ANALYTICS STACK

```
Copilot Studio Agent (existing)
  ↓
Built-in analytics → sessions, engagement, escalation rates
  ↓
Application Insights → custom telemetry events
  ↓
KQL queries → confidence trends, feedback ratios, regressions
  ↓
Power BI dashboard → 3 charts for stakeholders
  ↓
Power Automate → golden test alerts + feedback capture
```

Five components, in build order:

1. **Copilot Studio built-in analytics.** This is your Layer 1 (usage) out of the box. Enable it and review: session completion rate, engagement rate, topic triggering heatmap, and escalation rate. The key metric to watch is *resolution rate* — sessions that ended without escalation. If this number is high, the agent is answering questions. If it's low, users are bailing. No configuration needed — just open Analytics in Copilot Studio and start reviewing.

2. **Application Insights custom telemetry.** Connect Copilot Studio to Application Insights (Settings → Advanced — built-in integration). Then add custom events via Power Automate flows triggered by agent actions: on every generative answer, log `confidence_score`, `sources_used[]`, `response_length`, `topic_triggered`. On every user feedback event, log `feedback_type` (thumbs up/down), `session_id`, `query_text`. On every escalation, log `escalation_reason` and `last_agent_response`. This is your Layer 2 (quality) data pipeline.

3. **Power BI dashboard (3 charts).** Connect Power BI to Application Insights via the KQL data source connector. Build three charts: **Chart 1** — daily active users + sessions (line chart, 30-day trend). **Chart 2** — confidence score distribution (histogram, bucketed by topic). **Chart 3** — user satisfaction (thumbs up/down ratio, rolling 7-day average). Set the data refresh to daily automatic. Share the dashboard with the agent owner and their leadership. This is what turns "some people like it" into "here's the data."

4. **Regression alert flow.** Create the golden test set: 20 question-answer pairs stored in a SharePoint list. Build a weekly Power Automate flow that sends each question to the agent via HTTP, captures the response, and compares it to the expected answer using Azure OpenAI similarity scoring. If similarity drops below your threshold (start at 0.8), fire a Teams alert to the agent owner: "3 of 20 golden test answers degraded this week — review recent knowledge changes." This is the regression canary from Issue #016, fully automated.

5. **User feedback capture.** Add an Adaptive Card at the end of each agent response: "Was this helpful? 👍 👎" Build a Power Automate flow that logs feedback to a Dataverse `UserFeedback` table. Generate a weekly digest: Top 5 negative-feedback queries → review and improve. Optional enhancement: on 👎, show a "Tell me more" follow-up that captures qualitative feedback. This closes the loop between user experience and agent improvement.

> **Key insight:** The Power BI dashboard is what gets you the expansion meeting. Leadership doesn't read App Insights logs. But a clean three-chart dashboard showing adoption trending up, confidence holding steady, and user satisfaction at 85% — that's the artifact that turns a pilot into a program. Build the dashboard *first*, then fill it with data as you add each component.

STARTER KQL QUERIES FOR APP INSIGHTS

Sessions/day (30-day trend):

```
customEvents | where name == "AgentSession"
| summarize count() by bin(timestamp, 1d)
```

```
| render timechart
```

Avg confidence by topic:

```
customEvents | where name == "GenerativeAnswer"
| extend conf =
todouble(customDimensions.confidence_score)
| summarize avg(conf) by
tostring(customDimensions.topic_triggered)
```

Feedback ratio (7-day rolling):

```
customEvents | where name == "UserFeedback"
| summarize up=countif(customDimensions.feedback_type ==
"up"),
  down=countif(customDimensions.feedback_type == "down")
by bin(timestamp, 1d)
| extend ratio = round(todouble(up) / (up + down) * 100,
1)
```

⚠ **Similarity scoring threshold:** Start at 0.8 for the golden test set. Too high (0.95+) and every minor wording change triggers a false alarm. Too low (0.6) and real regressions slip through. Calibrate by running the test set twice against your current agent — if the same question produces similarity of 0.85–0.95 across runs, set your alert threshold just below the lowest score. Expect to tune this in the first two weeks.

🛠️ **TRY THIS NOW**

Add Analytics to Any Agent in One Session

1. **Enable Application Insights in your Copilot Studio agent.** Go to Settings → Advanced and connect your App Insights instance. If the customer doesn't have one, create it in the same resource group as their other Azure resources — it takes 30 seconds. This immediately starts flowing session telemetry into a queryable data store.

2. **Add a thumbs-up/down Adaptive Card after generative answers.** In the agent's generative answers topic, add a "Send a

message" node after the response with an Adaptive Card containing two buttons: 👍 and 👎. Connect each button to a Power Automate flow that logs the feedback type, session ID, and original query to a Dataverse table. This takes 15 minutes and gives you the user satisfaction signal immediately.

3. **Build 3 KQL queries in Application Insights.** Open the App Insights Logs blade. Write three queries: sessions per day (30-day trend), average confidence score by topic, and feedback up/down ratio. Use the starter KQL from the Pattern Breakdown above. Save each query — you'll pin them to Power BI in the next step.

4. **Pin them to a Power BI dashboard and share with the agent owner.** From each KQL query result, click "Pin to dashboard" → select or create a Power BI dashboard. Set the refresh to daily. Share the dashboard link with the agent owner and their leadership. You now have a live, auto-refreshing answer to: "How is the agent doing?"

💡 **The line that opens doors:** "Right now, can you tell me how many people used your agent yesterday and whether the answers were good? This dashboard answers both in real-time."

Got an agent or pattern to share?

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